



COURSE DESCRIPTION FORM: CS-2004: Fundamentals of Software Engineering

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

PROGRAM(s) TO BE EVALUATED BS-DS: Fall-2023

Course Description

Course Code	CS-2004	
Course Title	Fundamentals of Software Engineering	
Credit Hours	3	
Course Instructors	Ansa Liaqat, Laraib Afzaal	
Grading Policy	Absolute Grading	
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e., late submission will not be accepted. For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.	
Course Plagiarism Policy	Plagiarism in project or sessional/ final exam will result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.	
Prerequisites by Course(s) or Topics		
Assessment Instruments with Weights (homework, quizzes, sessional exams, final exam, assignments, etc.)	Assessment with the weight.	
	Assessment Type	Weight
	Quiz and in-class activities	15
	Sessional Exams 1	12.5
	Sessional Exams 2	12.5
	Project	20
	Final Exam	40
Course Coordinator	Ansa Liaqat	
URL (if any)		
Course Catalog Description	This course introduces students to the Overview of Software Engineering, Professional software development, Software engineering practice, Software process structure, Software process models, Agile software Development, Agile process models and Agile development techniques. Course will also cover Requirements Engineering, Design, architecture, testing, and project management in detail.	

Textbook(s)	Software Engineering (10th Edition) by Sommerville, Ian Addison Wesley		
Reference Material	Software Engineering: A Practitioner’s Approach, Pressman, R.S. & Maxim B., 8th Edition (2015), McGraw-Hill.		
Course Goals	A. Course Learning Outcomes (CLOs)		
	After course completion, the students shall be able to: 1. Students will learn the basics of software process models and will learn to choose the best fit for variable nature of projects in industry. 2. Discover both functional and non-functional requirements for a medium sized software system. 3. Students will learn to design and test their software. 4. Students will able to choose the appropriate architecture for their projects. 5. Students will be able to learn basic project management skills along with costing and estimation. 6. Students will learn to carry out a medium size project from scratch to end on their own using Iterative method.		
	B. Program Learning Outcomes (PLOs)		
	PLO 1	Computing and Artificial Intelligence Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to solve complex computing problems using artificial intelligence techniques.
	PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computational problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, computing, and artificial intelligence.
	PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
	PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods
	PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing and artificial intelligence tools, including prediction and modelling for complex computing problems.
	PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.

	PLO 7	Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing and artificial intelligence work in solving complex computing problems.											
	PLO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing and artificial intelligence practice.											
	PLO 9	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.											
	PLO 10	Communication	Communicate effectively on complex computing and AI activities with the computing and artificial intelligence community and with society at large.											
	PLO 11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.											
	PLO 12	Life Long Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.											
	C. Mapping of CLOs to PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)													
		PLOs												
		1	2	3	4	5	6	7	8	9	10	11	12	
CLOs	1	✓	✓	✓	✓					✓	✓		✓	
	2	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	
	3	✓	✓	✓	✓	✓							✓	
	4		✓	✓	✓	✓								
	5	✓	✓	✓					✓	✓	✓	✓	✓	
	6	✓	✓	✓		✓				✓	✓	✓	✓	
Topics covered in the course	Topics to be covered:													

(assume 15-week instruction and 3 contact hours per week)	List of Topics		No. of Weeks	Contact Hours	CLO(s)
	Introduction to Software Engineering		1	3	1
	Software Process Models (traditional models)		2	6	1,2,6
	Software Process Models (non-traditional models)		2	6	1,2,6
	Requirement Engineering		2	6	2
	Software Architecture design		2.5	7.5	1,3,4,6
	Software Testing Basics		2	6	1,3,6
	Software Quality Assurance and Processes		1.5	4.5	3
	Software Project Management (Planning, Costing, Scheduling)		2	6	1,5
	Total		15	45	
Programming Language for Assignments (if any)	C++, JAVA, Python				
Class Time Spent (in percentage)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues	
	50	25	20	5	
Oral and Written Communications	Every student is required to submit at least __4__ written reports of typically __4__ pages and make __1__ oral presentation of typically __10__ minutes' duration.				